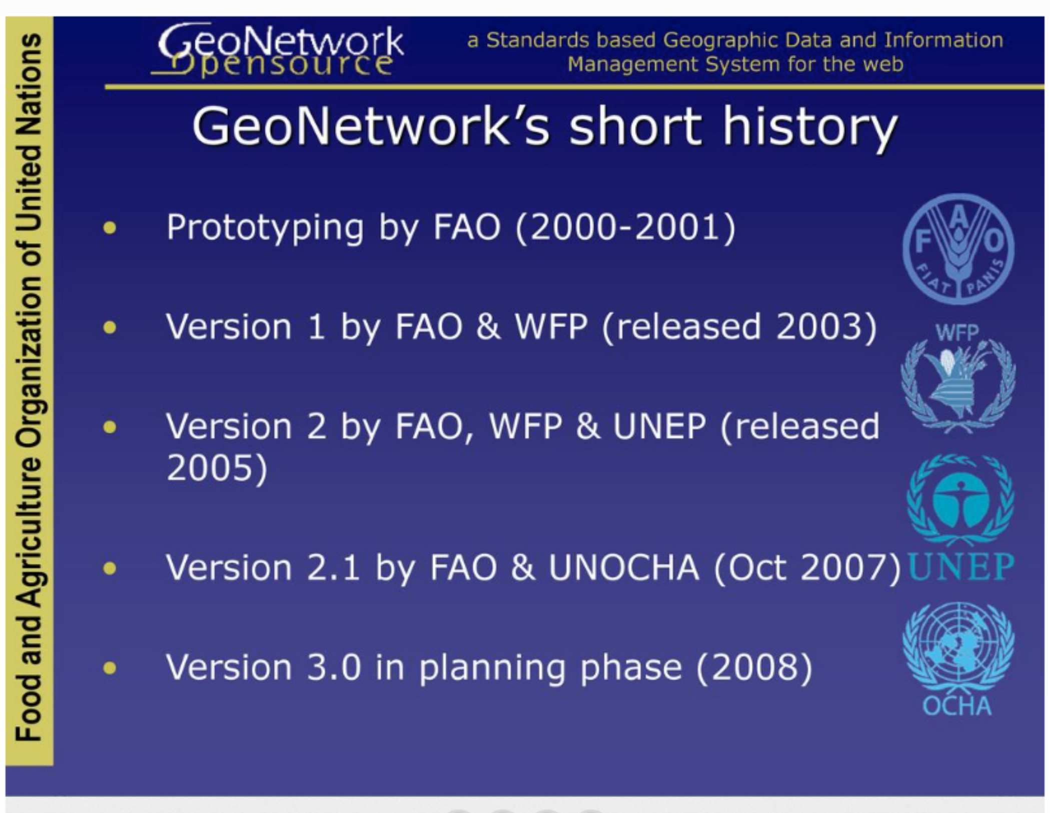

GeoNetwork OpenSource: Geographic data sharing for everyone

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1 GeoNetwork OpenSource: Geographic data sharing for everyone

Welcome to FAO/NRCE
GeoNetwork is a Geographic Information Management System that enables easy and timely access to spatial information and cartographic products through descriptive metadata from a variety of data providers using the capacities of the Internet.

2 GeoNetwork Main Goals

Provide a common platform and standards to manage geographic data
Improve accessibility and sharing of a wide variety of geographic data at different scale and from multidisciplinary sources
Increase collaboration between UN and other international organizations for collecting data and make them available to the internet community
The main objectives of the project are:
to provide a common platform and standards to manage geographic data
to improve accessibility to data and information
to increase collaboration within and between organization for reducing duplication and enhancing information consistency and quality and data preparedness.

3 GeoNetwork's short history

Prototyping by FAO ( )
Version 1 by FAO & WFP (released 2003)
Version 2 by FAO, WFP & UNEP (released 2005)
Version 2.1 by FAO & UNOCHA (Oct 2007)
Version 3.0 in planning phase (2008)
The prototype catalog system was developed in FAO in 2001 to systematically archive and publish the geographic datasets produced within the Organization.
Then WFP joined the project and with its contribution the first version of the software was released in 2003.
At that moment it was decided to develop the software as a Free and Open Source Software to allow the whole geospatial users community to benefit from the development results.
With UN Environmental Program (UNEP) a second version was released in 2005.
Now we are close to the latest version with the contribution of FAO and OCHA and the active support of CSI-CGIAR and UNEP
The version 3 is planned for next year

4 OGC Portal Reference Architecture

GeoNetwork has been developed based on the principles of a Free and Open Source Software and International Standards for services and protocols, like the ISO-TC211 and the OGC specifications. Its architecture is largely compatible with the OGC Portal Reference Architecture which is the OGC guide for implementing a standardized geospatial portal.

5 Infrastructure Web Interface

GEONETWORK is an Internet Spatial Data Platform structured in three main modules
Web Interface
Metadata
Catalogue
The software's structure in fact relies on three main modules focusing on the GIS community needs:
the Metadata-Module
Spatial data Repository
Dynamic Map-Viewer
All of them accessible through a web interface
provides
- an easy online metadata editor
- standard profiles for describing geodata
- data access through direct link to the data location
the Data-Module which is a Spatial data Repository
facilitates
- timely data storage and exchange
- organization and management of geodata
the Dynamic Mapping-Intermap
enables
- data visualization
- map production through query of multiple sources
Spatial Data
Repository

6 GeoNetwork Main Functions

Metadata and data publication and distribution
Metadata and data search capabilities in distributed environment
Interactive access to maps
Administrative functions
Metadata editing and management
Different metadata standards
Different sharing levels
This slide summarizes the main functionalities of the software;
first of all the metadata and data publication and distribution, then the metadata search capabilities in distributed environment (I will explain later why the search can be distributed)
Another function is the visualization of the data on line,
the Administration functions to manage metadata and users,
the metadata editing and management using different standards and different levels of sharing

7 The Basic Building Blocks

ISO TC211 Standards
OGC Standards (Open Geospatial Consortium)
We said the software is based on Standards:
Standards as support for describing data and standards for searching and retrieving information from catalogs.
STANDARDS for VISUALIZING DATA. The web viewer integrated in GEONETWORK (INTERMAP) in fact is compliant with the OGC WMS and ArcIMS map services standards.

8 Metadata Standards Support for multiple metadata standards

ISO19115:2003
FGDC
Dublin Core
GeoNetwork supports the most common standards to specifically describe geographic data (ISO and FGDC) and the international standard for general documents (Dublin Core)

9 Standard Protocol for Catalog

Full support of the OGC Catalog Services for the Web (CSW)
It also fully compliant with the OGC specifications in the way it queries and retrieves information to and out of a web catalog

10 Metadata Harvesting and Synchronization

I would like to emphasize the role of the metadata harvesting within the concept of data sharing.
Through this functionality it is possible to collect information from the different GNW nodes installed around the world and to copy and store periodically this information locally. In this way a user from a single entry point can get information also from distributed databases.

11 GeoNetwork Installations around the World

UN Organizations:
FAO, WFP, UNEP, WHO, OCHA, UNHCR
Other:
CGIAR, ICIMOD, ESA, GMFS, JRC, FGDC
Individual projects in Spain, France, Italy, Czech Republic, Australia, South Africa, ...
THIS IS THE web interface of the GEONETWORK OS SOFTWARE
and some examples of customization of the site in FAO, WFP and OCHA in NY.

12 The Home Page

It also fully compliant with the OGC specifications in the way it queries and retrieves information to and out of a web catalog

13 The Default Search

It also fully compliant with the OGC specifications in the way it queries and retrieves information to and out of a web catalog

14 The Search Results

It also fully compliant with the OGC specifications in the way it queries and retrieves information to and out of a web catalog

15 Display Metadata

It also fully compliant with the OGC specifications in the way it queries and retrieves information to and out of a web catalog

16 Display Metadata

It also fully compliant with the OGC specifications in the way it queries and retrieves information to and out of a web catalog

17 Display Interactive Map

Display Metadata
It also fully compliant with the OGC specifications in the way it queries and retrieves information to and out of a web catalog

18 Open Map Viewer

It also fully compliant with the OGC specifications in the way it queries and retrieves information to and out of a web catalog

19 Adding a Layer

It also fully compliant with the OGC specifications in the way it queries and retrieves information to and out of a web catalog

20 Exporting a Map

It also fully compliant with the OGC specifications in the way it queries and retrieves information to and out of a web catalog

21 Map Layout

It also fully compliant with the OGC specifications in the way it queries and retrieves information to and out of a web catalog

22 The Advanced Search

It also fully compliant with the OGC specifications in the way it queries and retrieves information to and out of a web catalog

23 Food and Agriculture Organization of United Nations

Thanks
Patrizia Monteduro
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